A Multi-Objective Optimization Technique for Volt-Var Control with High PV Penetration using Genetic Algorithm

Temitayo O Olowu, *Student Member, IEEE,* Mohamadsaleh Jafari, *Student Member, IEEE,*
Arif I Sarwat, *Senior Member, IEEE*, Department of Electrical and Computer Engineering,
Florida International University, United States.
asarwat@fiu.edu

***Abstract*—Voltage fluctuation issue is one of the primary challenges occurring with increasing PV penetration. Traditionally, voltage optimization was done with the use of Load Tap-Changing (LTC) transformers and capacitors. IEEE 1547a as amended in 2014 now allows the use of smart inverters on the distribution network by utility companies. This paper presents a co-optimization technique using Genetic Algorithm (GA) that uses an individual optimization algorithm of LTCs, shunt capacitors and smart inverters to determine the optimal settings for the devices. The PV smart inverters were set in the power factor control mode. The proposed technique was implemented on the IEEE 34 bus distribution feeder. The simulation results showed a reduction in the overall system power losses, minimized reactive power injection, optimal smart inverter power factor settings and ideal conservative voltage reduction.**

***Index Terms*—Volt-VAR control, Optimization, Genetic Algorithm, High PV penetration**

NOMENCLATURE

CVR	Conservative Voltage Reduction
j	Node Index
N_c	Number of Shunt Capacitors
N_n	Number of Nodes
P_{loss}	Total System Active Power Losses
Q_{cbmax}	Maximum Capacitor kVAR
Q_{cb}	Total System Reactive power Injection by Shunt Capacitor
Q_{loss}	Total System Reactive Power Losses
S_{loss}	Total Apparent Power Losses
t	Hourly Time Index
Tap_{vr}	Tap Position of Voltage Regulators
$V_{j,t}$	Actual Voltage of Node j at Time t
V_j^{max}	Maximum Voltage at Node j
V_j^{min}	Minimum Voltage at Node j
VAR	Reactive Power
VVO	Volt-VAR Optimization

I. INTRODUCTION

THE growing concern for global warming has propelled many utility and power generation companies to look for alternative and environmentally friendly sources of power generation [1–3]. The proliferation of both rooftop and utility-scale PV systems have increased in recent years. The reduction in the cost of installations, increase in the efficiency of solar panels as well as environmental benefits have led to a rapid growth in the use of rooftop PV systems in the United States [4]. A report by [5] showed that there was a 50% increase in PV capacity installations around the world in 2016.

The challenges associated with increasing use of grid-tied PV systems are caused by their intermittent nature [6]. The feeder voltage is usually impacted by this intermittency [7]. Several methods have been proposed by various researchers on how to mitigate these challenges. These ranges from the use of energy storages [8], switched capacitor banks [9], use of dump loads for active power curtailment [10], smart inverters [11] to use of FACT devices such as STATCOMs, SVCs etc. for reactive power compensation [12]. For utility-scale PV systems connected to the distribution feeders, increase in the penetration levels of PV systems leads to incessant operation of legacy devices such as substation Load Tap-Changing (LTC) transformers, substation feeder or bus voltage regulators, line voltage regulators and fixed and switched capacitors. This consequently affects the life span of these devices as presented in [13, 14].

Traditionally, LTCs with Voltage Regulators (VRs) and later capacitors were used by utility companies for voltage regulation. The LTCs basically switch the tapping of the transformers either up or down to effect the voltage of the feeders within the ANSI C84.1 standard range. The capacitors are used for reactive power injection which consequently leads to voltage control. Several studies have been done on the coordination of these two devices for optimal use for voltage regulation [15, 16]. Prior to the amendment made to the IEEE 1547a standard in 2014, Distributed Energy Resources (DERs) smart inverters were not permitted to participate in the distribution feeder voltage regulation. With the improvement of the smart inverter technologies, the amendment now allows the use of DERs such as PV systems to participate in the voltage regulation with other legacy devices such as On Load Tap Changing (OLTC) and capacitors [17].

Volt-VAR Optimization (VVO) control can be done at two levels in the distribution network. The first level is being

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the centralized VVO (or network-based VVO). This level of VVO is done at a centralized location that is integrated in the Distribution Management System (DMS). The main challenge of this system is the ability to process large data and provide VVO solutions in real or quasi-real time.

The decentralized level allows VVO to be done on the devices located on the feeders. This paper is based on VVO at a decentralized level. The optimal control of these three devices is very important for smooth and economic operation of the whole system. Since these three devices have the capacity of adjusting the voltage profile of the feeders differently, therefore there is a need for optimal coordination within these three devices. This paper presents a multi-objective formulation for the CVR by making the bus voltage profile close to 0.95 pu as much as possible, minimizing the overall system losses as well as minimizing the reactive power injection by the shunt capacitor banks to the lowest level.

II. RELATED WORKS

Optimal scheduling and control of various voltage regulating devices has been proposed by different papers. Agus [18] proposed the use of evolutionary based algorithms to schedule LTCs and shunt capacitors. Ref [16] developed a coordinate control method for LTCs and capacitor banks to ensure a better voltage profile in a distribution feeder using both LTCs and capacitor banks at the same time by making some mathematical formulations. Iman et al [19] developed the optimal rating and also location of fixed and switched capacitors by using modified discrete particle swarm optimization. Reference [20] developed an optimal dispatch algorithm for LTCs and shunt capacitors with plug-in electric vehicles. A report published by NREL and subsequent publication [21, 22] showed the use of these capacitors, LTCs and smart inverters for voltage regulation and also analyzed the power quality effects of the co-optimization algorithm developed. The use of CVR was also implemented to see how much energy is saved by applying CVR with different levels of PV penetration.

III. PROPOSED OPTIMIZATION ALGORITHMS

The proposed algorithm optimizes the bus voltage profile to the 0.95 pu boundary to take some advantage of CVR, minimize the overall system losses as well as uses the least amount of reactive power from the capacitor banks. The losses considered in this paper are purely technical losses such as feeder losses, transformer load and no-load losses and capacitor bank losses. Some non-technical losses could include data theft (in case of smart grid network) and power theft, looses due to breakdown of malfunction of devices in the distribution network. Non technical losses are not considered in this work

A. Multi-Objective Formulation

1) Objective Function 1: The first objective of this work is to flatten the voltage profile of each feeder as much as possible close to the lower 0.95 pu threshold. This allows for the least amount of power consumption in the network which is basically the concept of CVR. Many load types has their power consumption reduced with a reduction in their terminal voltages. This Objective Function (OF) was formulated as eqution 1 similar to the expression derived by [23]

$$OF_1 = \min \sum_{j=1}^{N_n} \left| \left((V_j^{min} - V_{j,t}) + \left| \alpha V_{j,t} - V_j^{max} \right| \right) \right| \quad (1)$$

where V_j^{min} and V_j^{max} are the voltage constraints which are expressed in 4 and α is a weighting factor that was determined by making $V_{j,t}$ close to V_j^{min} as much as possible.

2) Objective Function 2: The second objective of the proposed method is to minimize the losses on each feeder within the time of 24 hour period of simulation. The load flow is solved on hourly load curve basis with a one-hour resolution PV data with the objective of minimizing the total system losses within these intervals.

$$\begin{aligned} S_{loss} &= \sqrt{P_{loss}^2 + Q_{loss}^2} \\ OF_2 &= min\ (S_{loss}) \end{aligned} \quad (2)$$

3) Objective Function 3: The third objective of this work is to minimize the total amount of reactive power injected by the two capacitor banks connected to buses 844 and 848.

$$OF_3 = min \sum_{cb=1}^{Nc} Q_{cb} \quad (3)$$

B. Constraints Formulation

Bus voltage constrains should be within the ANSI C84.1.

$$\begin{aligned} V_j^{max} \geq V_{j,t} \geq V_j^{min} \\ 1.05pu \geq V_{j,t} \geq 0.95pu \end{aligned} \quad (4)$$

Capacitor bank VAR and switching constrains is given as equation 5. The maximum reactive power injection by these capacitors are 300 kVAR and 450 kVAR according to the values in the feeder parameters.

$$|Q_{cbmax}^j| \geq |Q_{cb}^j| \quad (5)$$

Considering a voltage regulator with 32 tap steps, the constraints for the tap setting is given as

$$-16 \leq Tap_{vr,t} \leq +16 \quad (6)$$

The power factors of the six PV systems where set as variable in the optimization algorithm to get the optimal values of the PV power factors.

$$0.8 \leq PV_{PF} \leq 1 \quad (7)$$

IV. TEST FEEDER AND SIMULATION

The test feeder is a IEEE 34 node test feeder with a nominal voltage of 24.9 kV as shown in figure 1. The feeder has two units of 3-phase voltage regulators places between line segments 814-850 and 852-832. The system has a 2.5MVA, 69 / 24.9 kV, $\Delta - Y$ substation transformer and 0.5MVA, 24.9 / 4.16 kV, Y-Y transformer connected. There are two 3-phase

shunt capacitors of 300 kVAR and 450 kVAR connected to nodes 844 and 848 respectively. 6 units of 300 kW solar PV panels were connected to nodes 822, 840, 846, 848, 864 and 890.

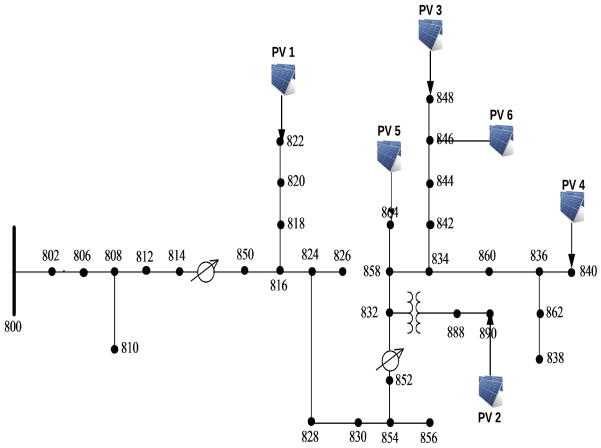


Fig. 1: IEEE 34 node test feeder with PV locations

Table I shows the specifications of the PVs located on the network. The default PF used for the non-optimized solution was 0.8. This is to enable the inverter inject some fixed reactive power into the system. The total kVA rating of the the load in the system is 2058 kVA. With a total PV inverter kVA of 1500 kVA, this constitute a PV penetration level of 70 % approximately using the traditional method of PV penetration estimation which give the amount of PV penetration as the ratio of the PV generation to the maximum load on the feeder.

TABLE I: PV Specifications

PVs	Maximum Power (kW)	Inverter (kVA)	Phases	Default PF
PV1 and PV5	300	250	1	0.8
PV2, PV3, PV4 and PV6	300	250	3	0.8

1) Simulation and Results: The simulation of the test feeder was done with OpenDSS software with the optimization done with MATLAB software. The time series analysis was done with a 24-hour duration. The PV irradiance was of 1-hour resolution with peak generation at 13th hour of the day.

The nodes of interest were the ones where the PVs were connected. The optimization was done on a hourly bases. Figures 2-3 show the single phase voltage profile of buses 822 and 864 while figures 4-7 show the 3-phase voltage on buses 840, 846, 848 and 890. The voltage profiles on these nodes showed significant reduction in the values in the terminal voltages without any violations. The voltage on node 890 was below the ANSI C84.1 voltage threshold of 0.95 without optimization. After optimization, the voltage was increased beyond the 0.95 pu threshold.

The plots of the losses in each hour of simulation is as shown in figures 9. The plots showed a consistent reduction in the system apparent losses. The active power losses was reduced in all the hours except the 14th, 17th and the 23rd hours. This obviously was due to need to keep the voltage within the required threshold. The values shown in table III are the voltages (both the optimal and the non-optimized) when there is maximum generation from the PV system at the 13th hour of simulation. Table II contains the optimal values of the power factor of the PV systems at the 13th of simulation. These values show some deviations from the 0.8 PF value set for the non-optimized solution. PV1 and PV6 had nearly unity PF which indicated almost no reactive power injection by these PVs while PV2 had the least PF, consequently with the maximum value of reactive power injection.

The bar charts in figure 8 shows the reactive power injection on nodes 844 and 848 by the capacitor banks. The optimized values of the kVAR injection showed a significant reduction compared to the simulation without optimization.

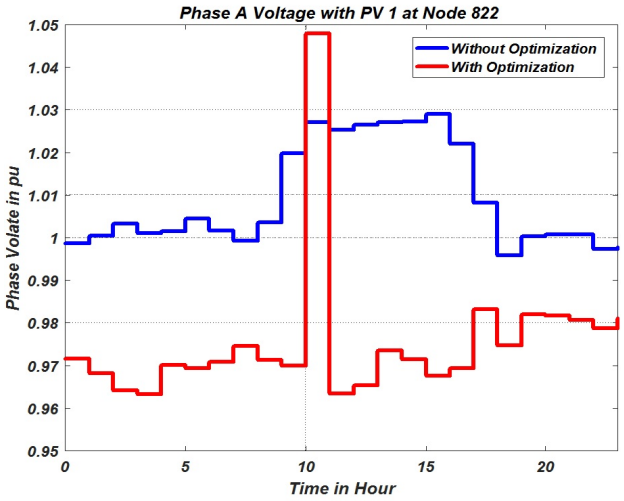


Fig. 2: Voltage Profile of Node 822

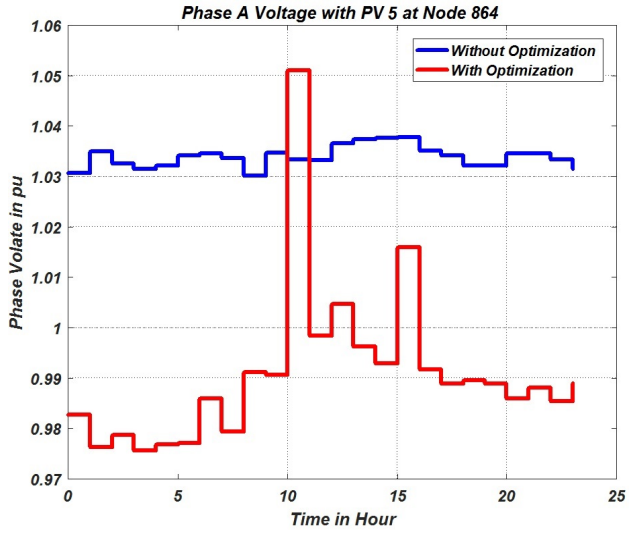


Fig. 3: Voltage Profile of Node 864

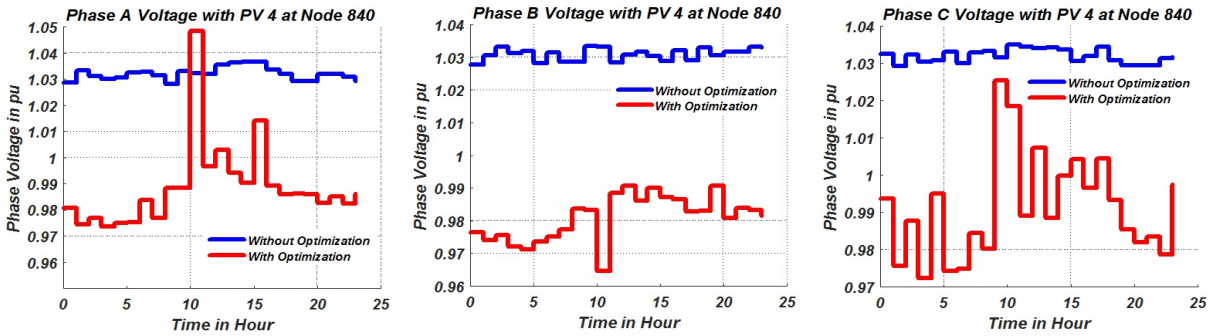


Fig. 4: Voltage Profile of Node 840

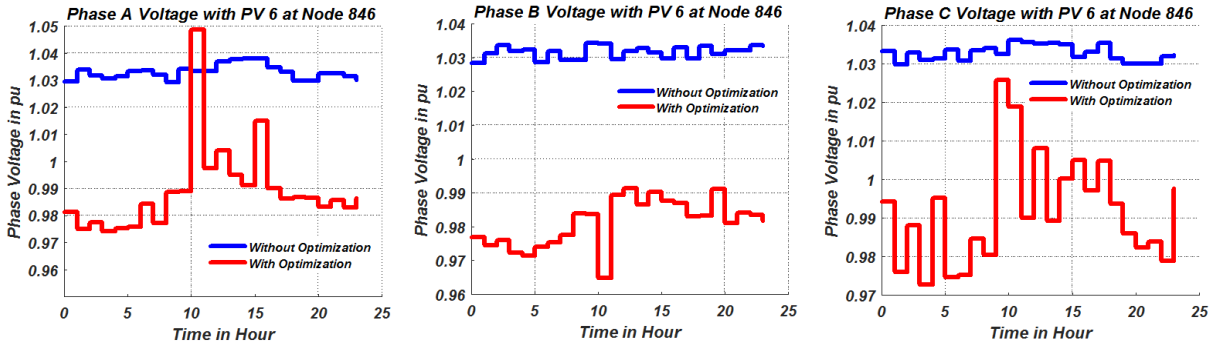


Fig. 5: Voltage Profile of Node 846

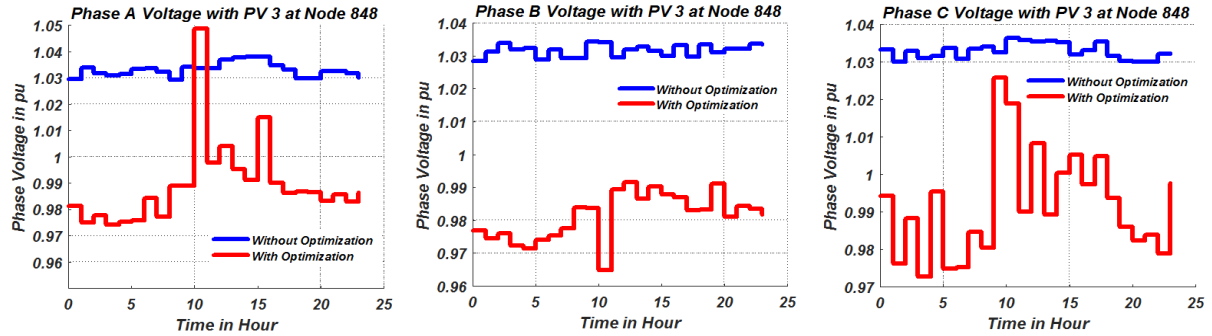


Fig. 6: Voltage Profile of Node 848

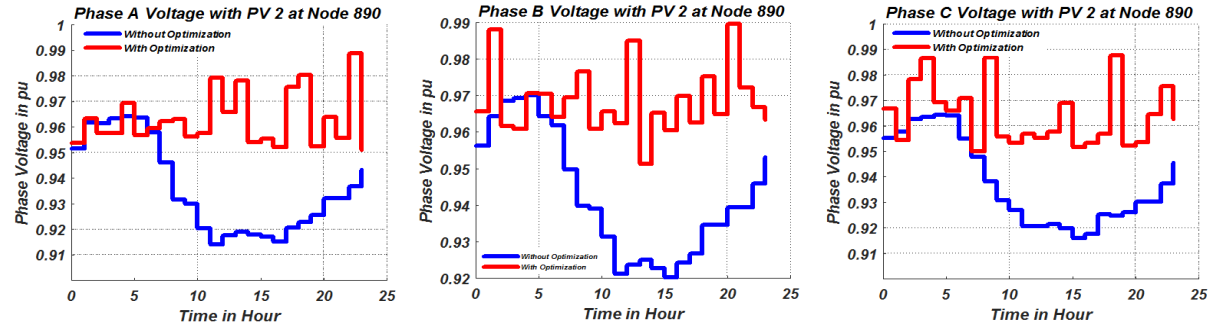


Fig. 7: Voltage Profile of Node 890

TABLE II: PV Power Factors Optimized

	PV1	PV2	PV3	PV4	PV5	PV6
PF	1	0.82	0.917	0.97	0.947	0.99

TABLE III: Node Voltages for the for all the nodes in the 13th hour of simulation

Bus	**Distance (km)**	**Va (Normal)**	**Va (Optimal)**	**Vb (Normal)**	**Vb (Optimal)**	**Vc (Normal)**	**Vc (Optimal)**
sourcebus	0	1.0500	1.0500	1.0500	1.0500	1.0500	1.0500
800	0	1.0500	1.0500	1.0500	1.0500	1.0500	1.0500
802	0.79	1.0503	1.0500	1.0486	1.0484	1.0496	1.0494
806	1.05	1.0505	1.0500	1.0478	1.0473	1.0493	1.0490
808	1.31	1.0543	1.0495	1.0326	1.0286	1.0453	1.0424
810	11.14	0	0	1.0324	1.0284	0	0
812	12.02	1.0584	1.0487	1.0159	1.0077	1.0404	1.0343
814	12.91	1.0615	1.0480	1.0028	0.9910	1.0367	1.0278
814r	22.57	1.0217	0.9563	1.0153	0.9973	1.0302	0.9828
850	31.63	1.0217	0.9563	1.0153	0.9973	1.0302	0.9828
816	31.63	1.0217	0.9563	1.0151	0.9970	1.0301	0.9827
818	31.63	1.0216	0.9563	0	0	0	0
824	31.73	1.0231	0.9557	1.0086	0.9894	1.0267	0.9785
820	32.25	1.0226	0.9605	0	0	0	0
822	33.28	1.0265	0.9654	0	0	0	0
826	34.84	0	0	1.0084	0.9893	0	0
828	39.59	1.0232	0.9557	1.0081	0.9889	1.0265	0.9781
830	46.92	1.0262	0.9548	0.9970	0.9756	1.0196	0.9694
854	49.02	1.0263	0.9548	0.9968	0.9753	1.0195	0.9692
832	51.11	1.0331	1.0023	1.0325	0.9937	1.0337	1.0082
858	35.30	1.0355	1.0037	1.0314	0.9921	1.0337	1.0078
834	35.76	1.0353	1.0029	1.0310	0.9909	1.0338	1.0071
842	34.97	1.0354	1.0029	1.0310	0.9909	1.0338	1.0071
836	35.09	1.0354	1.0028	1.0308	0.9907	1.0339	1.0072
840	38.21	1.0355	1.0029	1.0309	0.9908	1.0340	1.0073
844	41.32	1.0355	1.0029	1.0310	0.9908	1.0339	1.0071
846	41.48	1.0368	1.0040	1.0319	0.9915	1.0353	1.0082
848	52.71	1.0369	1.0040	1.0320	0.9915	1.0354	1.0083
856	53.46	0	0	0.9967	0.9752	0	0
852	54.21	1.0331	0.9546	0.9775	0.9520	1.0085	0.9546
852r	55.09	1.0331	1.0023	1.0325	0.9937	1.0338	1.0082
888	55.98	0.9975	0.9662	0.9971	0.9584	0.9997	0.9742
864	56.07	1.0366	1.0046	0	0	0	0
860	57.42	1.0352	1.0027	1.0308	0.9907	1.0337	1.0070
862	57.55	1.0354	1.0028	1.0308	0.9907	1.0339	1.0072
838	57.68	0	0	1.0306	0.9905	0	0
890	56.27	0.9175	0.8858	0.9237	0.8850	0.9207	0.8953

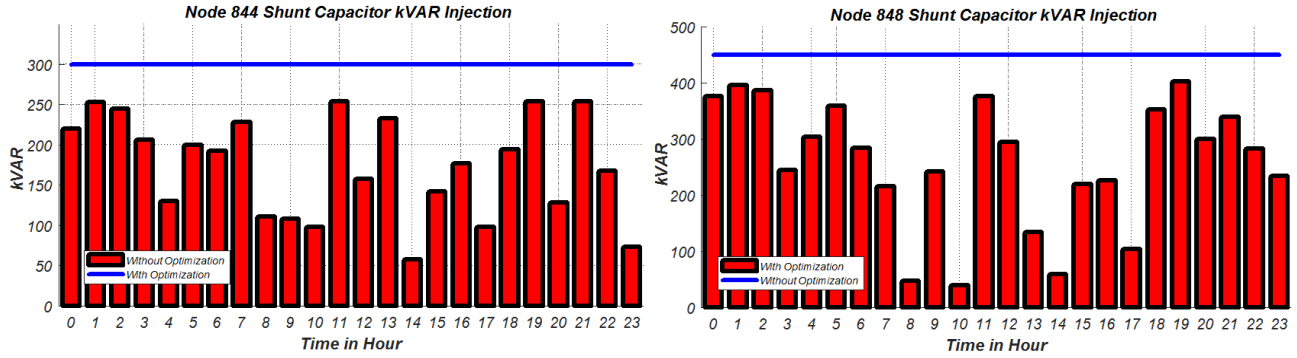


Fig. 8: kVAR injection on nodes 844 and 848 by shunt capacitors

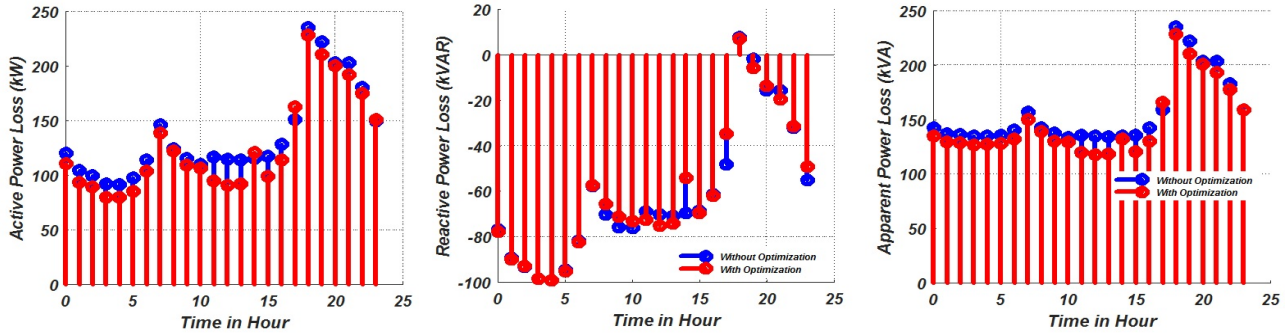


Fig. 9: Hourly losses in the system

V. Conclusion

This paper presented a multi-objective optimization technique for volt-var control with high PV penetration on the distribution feeder using genetic algorithm. This optimization was solved to minimize the distribution system losses while keeping the voltage of the nodes close to the lower threshold of the ANSI C84.1 standard. This was done to take the benefit of CVR into effect. The results shows the effectiveness of the technique in the use of CVR and also in minimizing the system losses. There are also some trade offs in keeping the voltage lower where the overall system losses was slightly increased beyond the normal value.

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